# JBOWFT: Jailbreaking Model Only With Fine-Tuning

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## 1 Introduction

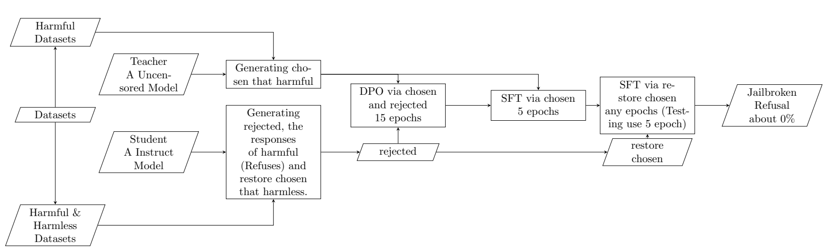
**Abstract**

Most Jailbreaking Method *(e.g. Heretic)* were based on Abliteration, it is a jailbreaking method via removing refusing directions of LLMs. But **JBOWFT** is different, it jailbreaks LLMs only via Fine-Tuning and Datasets.

We tested this method on a Instruct Model, **Qwen3.5-2B**, a strong safety model, after it fine-tuned via JBOWFT, it successfully outputs the harmful message, and in the *JailbreakHub* Datasets, it haven't any refusals. Therefore, it has been proven to be effective on this work.

## 2 How it works

It uses the Distillation-Jailbreaking such like **FLRT** (From *arXiv:2407.17447*). **FLRT** is finding a prompt that can jailbreak the model. But **JBOWFT** directly changes the default model weight via Fine-Tuning with Uncensored Teacher's Output and make it Uncensored.
General Process:




Basically, it averages the logits from chosen and rejected, then uses SFT to advance the logit of chosen. Finally, it uses another SFT to restore the basic capacities such as coding.

# 3 Result

## 3.1 Test Arguments

Model: **Qwen3.5-2B** *(Qwen/Qwen3.5-2B)*
Teacher Model: **Qwen3.5-9B-Uncensored** *(HauhauCS/Qwen3.5-9B-Uncensored-HauhauCS-Aggressive)*
Datasets: **mlabonne/harmful_behaviors** (the first 99 prompts) and **mlabonne/harmless_alpaca** (the first 25 prompts)
LoRA Rank: 512
LoRA Alpha: 1024 ($512 \times 2$)
Quantization: 4-bit
Max Sequence Length: 2048
DPO LR: $5 \times 10^{-5}$
SFT LR: $2 \times 10^{-4}$
DPO Beta: 0.05

## 3.2 Benchmark (OOD)

Judge: **Qwen3.5-9B-Uncensored** *(HauhauCS/Qwen3.5-9B-Uncensored-HauhauCS-Aggressive)*
Benchmark Prompts: **JailbreakHub** forbidden_question_set.csv (All 390 prompts)
Temperature: 0.0
Refusals: 0/390 (0.00%, Not included Soft Refusals)
Soft Refusals: 5/390 (1.28%)

## 3.3 KL Divergence

Harmless Prompt: From Test Application
LogProbs: Top 50 tokens
Mean KL: 0.690416
Std KL: 0.435010
Min KL: 0.176953
Max KL: 1.779243
This means the base ability was kept by **SFT Restore**, and the **Max KL Divergence** is not too high.

# 4 Discussion

## 4.1 Who is better

**JBOWFT** is effective, reached 0% refusals on OOD datasets. Anyways, **KL Divergence** (0.5-0.7) of this method is more than Abliteration (0.03-0.3). But the ASR of **JBOWFT** is better than most Fine-Tuning jailbreak via DPO or

SFT. This method improved the **ASR** of **ONLY** DPO and SFT jailbreaking, and if you want a low **KL Divergence**, your priority selection still is Abliteration. If you want a plain SFT or DPO jailbreaking, your priority selection is this.

### 4.2 Why KL Divergence is high

I guess, the **DPO** (Direct Preference Optimization) with beta=0.05 is the real culprit for increasing KL divergence. But I cannot remove **DPO**, if I remove **DPO**, the Refusals might be lower than this.

### 4.3 Future

Reducing KL divergence is the next challenge. Possible solutions include:

- Using a larger beta value (e.g., 0.1–0.2) in DPO to limit preference updates.
- Expanding the Restore SFT dataset.
- Add more Learning Rate to restore SFT. (May not)

## 5 Limitations

In some special situations, the model will output **SOFT REFUSALS** that means Model will output ”I cannot provide xxx” and similar, then output the safety or neutral content.

## 6 Conclusion

**JBOWFT**, is a Fine-Tuning Only Jailbreaking to approach the effect of Abliteration. After the three-stage Fine-Tuning, we successfully got 0% refusals on *JailbreakHub Datasets* benchmark, and 1.28% soft refusals. The KL Divergence that 0.690416 is similiar of some result on Heretic Jailbreaking. Our method is simple, effective and automatic. And it can fully uncensor a Strong Safety Alignment model (e.g. Qwen Series, Gemma Series).

# 7 Appendix

## 7.1 Contributor

Tianbo Wang, 12-year-old, for the thought and code.
DeepSeek, for fixing the bugs of Test APP.

## 7.2 Test Application

Open-Source with MIT License.
Site: https://github.com/Win12Home/AntiRegular

# References

[1] T. Ben Thompson and Michael Sklar. FLRT: Fluent Student-Teacher Redteaming. *arXiv preprint arXiv:2407.17447*, 2024.